Current Progress and Next Steps

Completed Work

- Integrated SUMO, OMNeT++ and Veins.
- Implemented V2V/V2I communication and OBU-based heuristic filtering.
- Integrated RSU-based ML detection and trust management.
- Embedded trained ML models (Random Forest and Logistic Regression) as native C++ predictors in the RSU for real-time edge inference.
- Implemented approximately 15 attack scenarios aligned with VeReMi categories.
- Completed 14-feature extraction and CSV logging.
- Added trust-score updates and detection visualization.
- Implemented simulation and trust-state checkpointing.
- Set up a two-RSU simulation scenario.
- Implemented inter-RSU (I2I) communication backbone for periodic trust synchronization and urgent threat alerts.
- Built an interactive web dashboard for real-time model evaluation and data inspection.

Remaining Work

- Complete trust synchronization between RSUs.
- Handle duplicate, stale and conflicting trust updates.
- Implement Fog-based regional trust aggregation.
- Integrate SDN-based adaptive forwarding.
- Test different traffic densities and attack scenarios.
- Evaluate accuracy, precision, recall, F1-score, false-positive rate, latency and packet delivery ratio.
- Test scalability with more RSUs and larger network scenarios.
- Benchmark inference latency and resource footprint on physical edge hardware.